Resource cost analysis (preview of new section content)

The T_{total} analysis above only accounts for the final QPU sampling, not the cost of training the angles on quantum hardware itself. To investigate what would happen if both the angle training and sampling were performed on actual quantum hardware, we perform the following analysis.

For each candidate parameter setting,

$$T_{\text{proxy}} = t_{\text{preprocessing}} + NM\,t_{\text{shot}} + Q\,t_{\text{shot}}, \tag{1}$$

where N is the number of COBYLA iterations, the same K from T_{AS} above, or its cumulative sum across sub-depths for Interpolation, M is the number of shots used per COBYLA iteration, Q is the final number of sampling shots once the angles have been trained and fixed, the same Q already used in T_{QPU} above, and t_{shot} is the calibrated per-shot duration defined below. $t_{\text{preprocessing}}$ is zero for Fixed Angles and Linear Ramp, the measured transfer-lookup time for Parameter Transfer, and for Interpolation an added term for its recursive matrix-product-state evaluations, which the base NM term does not capture.

To determine the recommended strategy, we run these optimizations on MPS (Aer), each problem instance sampling a random subset of settings from $N \in \{10, 20, 40, 50, 60, 75, 80, 100, 150\}$ COBYLA iterations, $M \in \{10, 25, 50, 100, 200, 500, 1000\}$ shots per iteration, and $Q \in \{100, 200, 250, 500, 1000, 2500$ final shots for the trained methods, Fixed Angles*, Linear Ramp, and Interpolation*. Fixed Angles† and Parameter Transfer require no training and sample only over Q, up to 500 000 and 300 000 shots respectively. This covers a total of five strategies.

Running the full $N \times M$ training grid plus the final Q shots on `ibm_boston` for every strategy and depth is not practical, so we simulate that grid with the MPS (Aer) evaluator of Sec. **??** as a hardware proxy and record its cost as T_{proxy}, then rescale the result to the duration `ibm_boston` would take, once assuming ideal noiseless shots and once with a noise correction. No hardware samples were collected for this analysis, only the resource axis is calibrated. We build these calibrated Window Stickers with the Stochastic Benchmark framework [**?**].

We calibrate t_{shot} two ways, both anchored to the hardware characterization in Sec. **??** and App. **??**. The noiseless calibration sets $t_{\text{shot}} = 1/2470\,\text{Hz} \approx 405\,\mu\text{s}$, the measured `ibm_boston` shot rate. The noise-corrected calibration instead sets $t_{\text{shot}} = t_{\text{shot}}^{\text{noiseless}}/\sqrt{\gamma}(p)$, where $\sqrt{\gamma}(p) = F_{CZ}^{N_{CZ}/2}$ is the fraction of a raw shot that constitutes one noiseless-equivalent sample, $N_{CZ} = 2 \times 143 \times p$ is the number of two-qubit gates for depth-p QAOA on the 144-qubit heavy-hex device, and $F_{CZ} = 99.872\%$ is the median `ibm_boston` CZ-gate fidelity. This penalizes deeper circuits more heavily. Table 1 lists the resulting per-shot durations and rescaling factors.

This analysis covers the strategies already shown on the Pareto frontier of Fig. **??**, Fixed Angles*, Fixed Angles†, Parameter Transfer, Linear Ramp, and Interpolation*. We ran the campaign of Sec. **??** for these families on 30 heavy-hex-144 MaxCut instances (20 training, 10 test), bootstrap-resampled into stochastic-benchmark statistics [**?**], fitting an actionable prescription curve per method on the training instances and evaluating it on held-out test instances. At each resource level we take the running maximum actionable response across families to obtain the Pareto frontier, crediting a method only where it has direct fitted evidence rather than flat extrapolation past its last measured point.

Figure 1 overlays the resulting actionable Pareto frontier under both calibrations on a single axis, so the effect of the hardware noise assumption on the recommended strategy is visible directly rather than split across separate figures. Line colour marks which method family holds the frontier at a given resource level; linestyle distinguishes the noiseless (solid) and noise-corrected (dashed) calibrations.

Table 1: Calibration factors for the noiseless and noise-corrected T_{proxy} resource axes, by depth p, measured on `ibm_boston`. s_{nl} and s_{nc} are the factors each method's raw T_{proxy} is rescaled by. Depth $p = 1$ has no local hardware timing records and is excluded.

p	t_{hw} (μs)	N_{CZ}	$\sqrt{\gamma}(p)$	t_{nl} (μs)	t_{nc} (μs)	s_{nl}	s_{nc}
2	271.27	572	0.6933	404.86	583.97	1.4925	2.1528
3	271.27	858	0.5773	404.86	701.35	1.4925	2.5855
4	271.27	1144	0.4806	404.86	842.33	1.4925	3.1051
5	280.31	1430	0.4002	404.86	1011.64	1.4443	3.6090
6	275.79	1716	0.3332	404.86	1214.98	1.4680	4.4055
7	298.39	2002	0.2775	404.86	1459.19	1.3568	4.8901
8	280.31	2288	0.2310	404.86	1752.49	1.4443	6.2520
9	271.27	2574	0.1924	404.86	2104.75	1.4925	7.7590
10	280.31	2860	0.1602	404.86	2527.81	1.4443	9.0179

At the smallest calibrated budgets, Fixed Angles†, without any angle reoptimization, holds the frontier under both calibrations, with occasional Parameter Transfer segments, consistent with these being among the fastest angle-setting methods in Fig. **??**. As the budget grows, Fixed Angles* takes over and holds the frontier across most of the remaining range, in particular the depth-six variant, which dominates the majority of the accessible budget on both panels. Interpolation* only claims the frontier in a narrow middle window before Fixed Angles* (p=6) reclaims it at the largest budgets, so its higher training cost does not translate into a lasting advantage once that training is itself priced onto the hardware axis, echoing the diminishing returns from extensive angle optimization already noted for Fig. **??**. Linear Ramp briefly holds the test-panel frontier at intermediate budgets under both calibrations but does not appear on the training-panel frontier in this sweep.

The noise-corrected calibration pushes the entire frontier to the right relative to the noiseless one, most visibly at the largest budgets, since deeper circuits lose a larger fraction of each shot to gate error and therefore need proportionally more of them to reach one noiseless-equivalent sample. The ordering of which family holds the frontier is otherwise preserved between the two calibrations, shallow unoptimized methods are cheapest at low budget, Fixed Angles* dominates the middle and upper range, and Interpolation* is only transiently competitive. The best achieved approximation ratio at the largest available budget is unchanged by the calibration choice (92.9% train, 91.9% test), since both calibrations rescale the same underlying best-performing configuration and only its apparent cost in duration shifts.

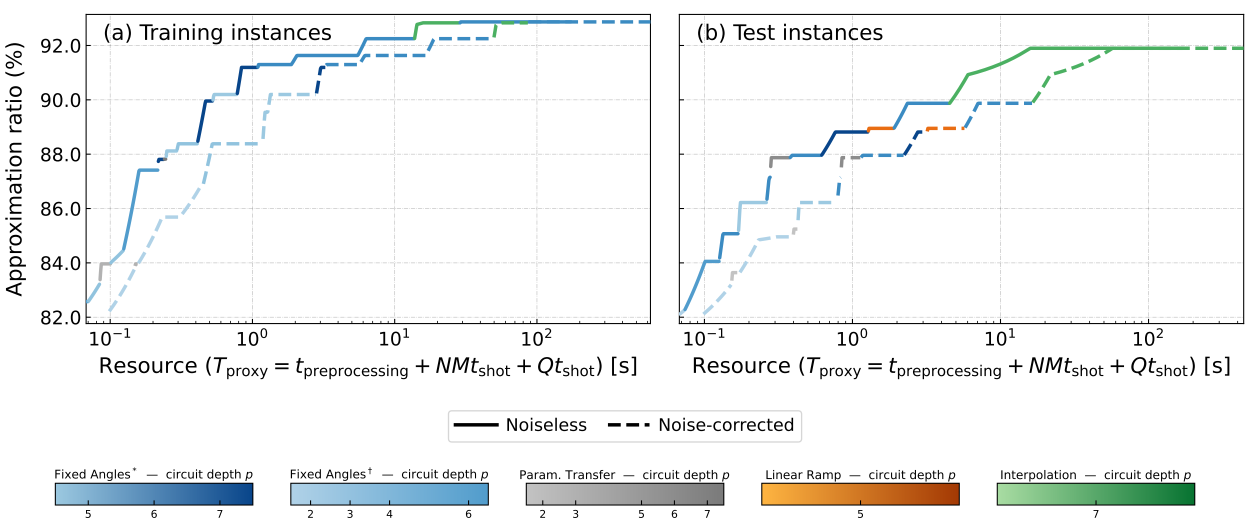

Figure 1: Actionable Pareto frontier for 144-node heavy-hex MaxCut under the noiseless and noise-corrected hardware calibrations (Eq. 1), overlaid on a single axis. (a) Training instances. (b) Test instances. Line colour indicates which method family is the best currently deployable strategy at that resource level. Solid lines are the noiseless calibration; dashed lines are the noise-corrected calibration.